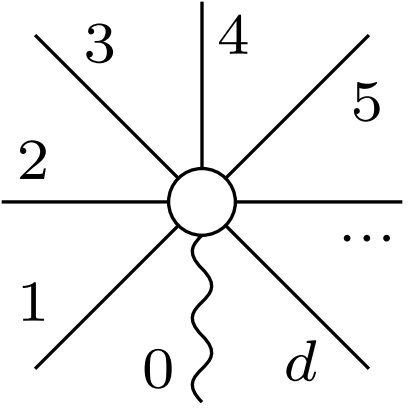
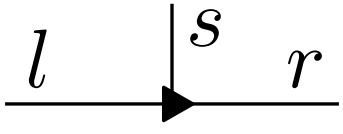
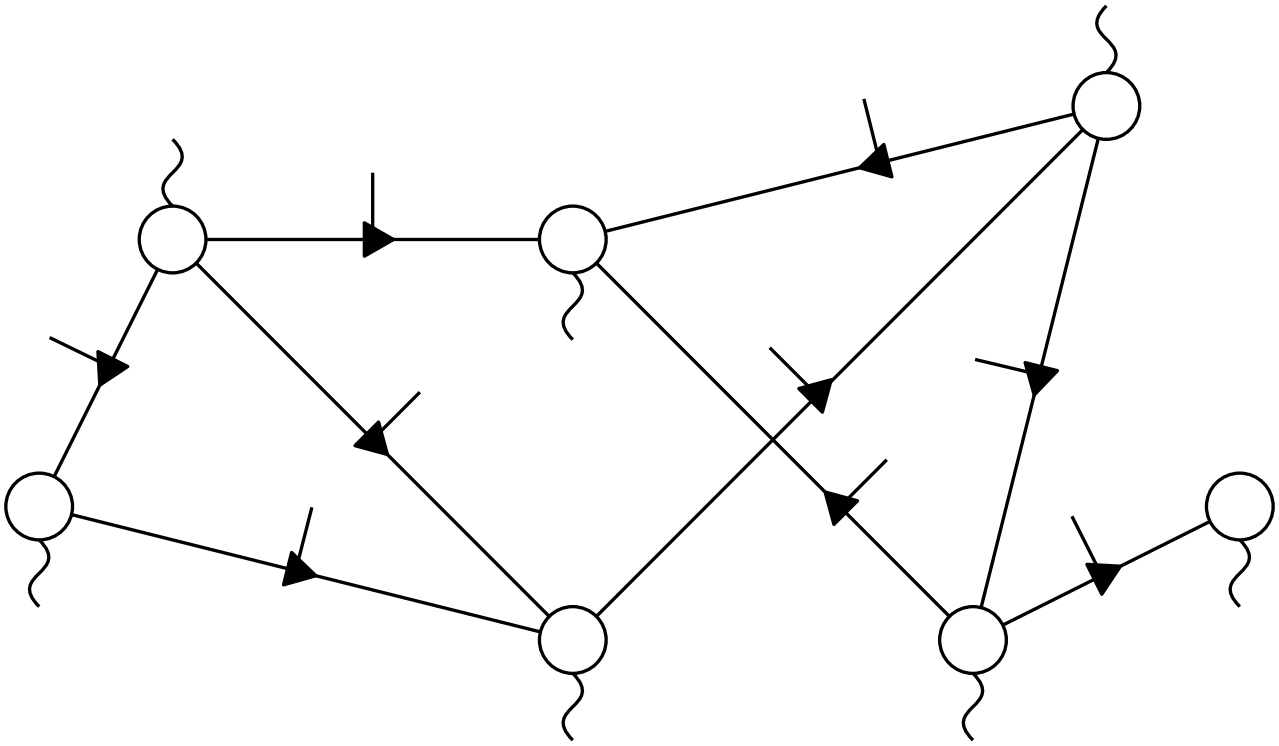
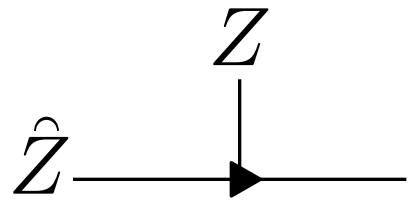

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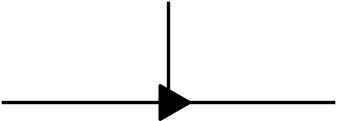

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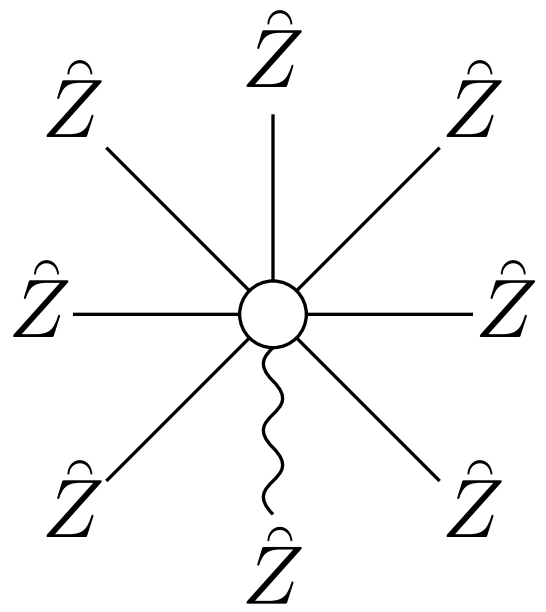


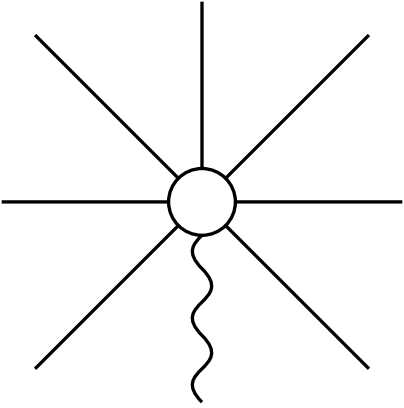

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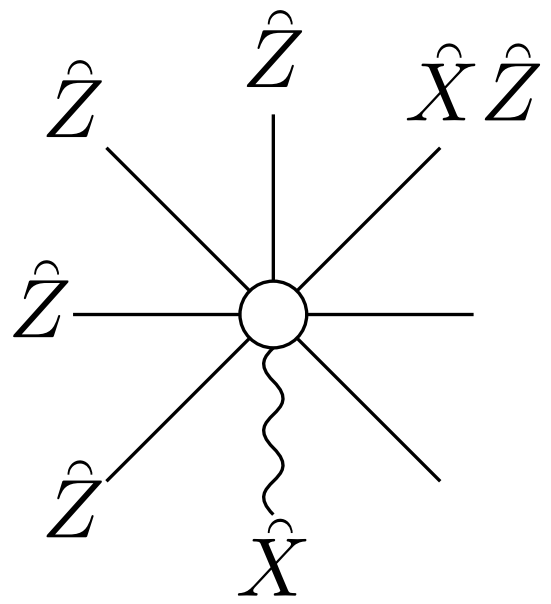

Z
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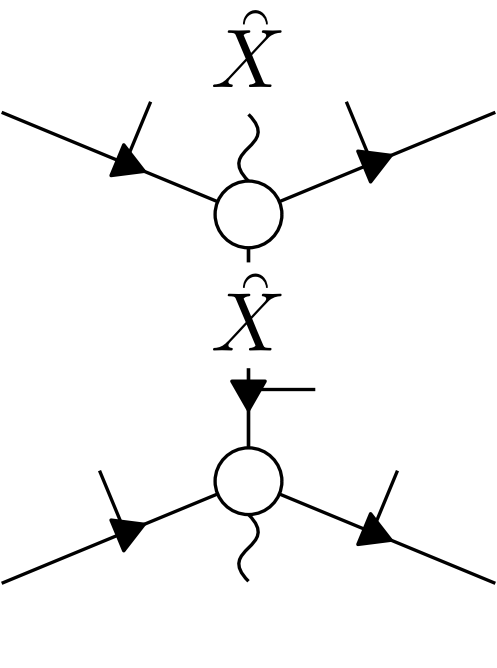

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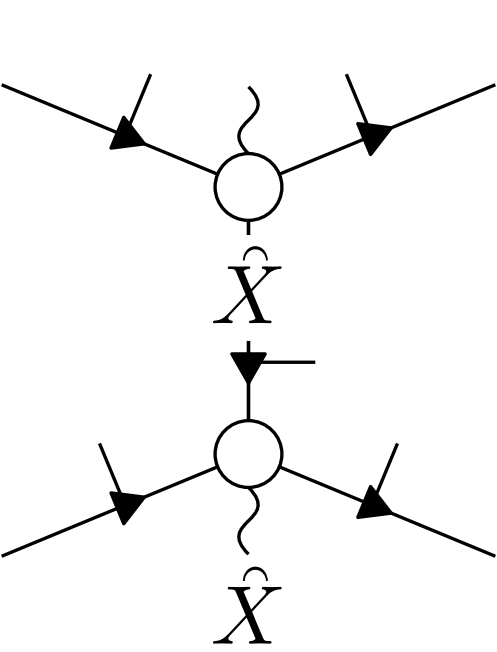


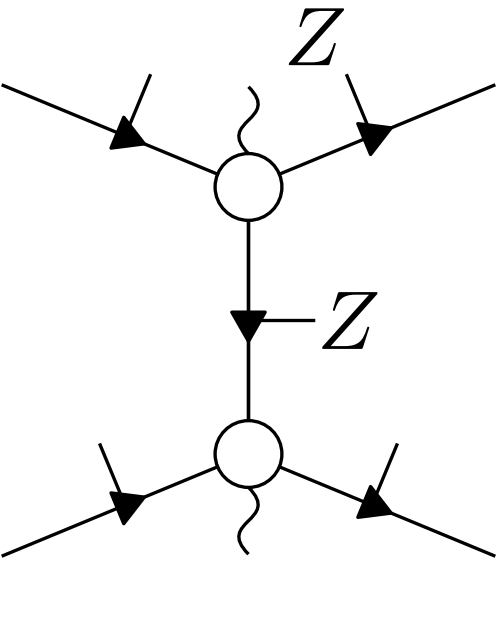

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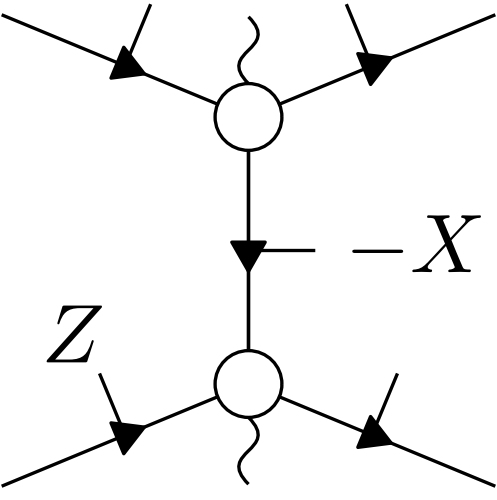


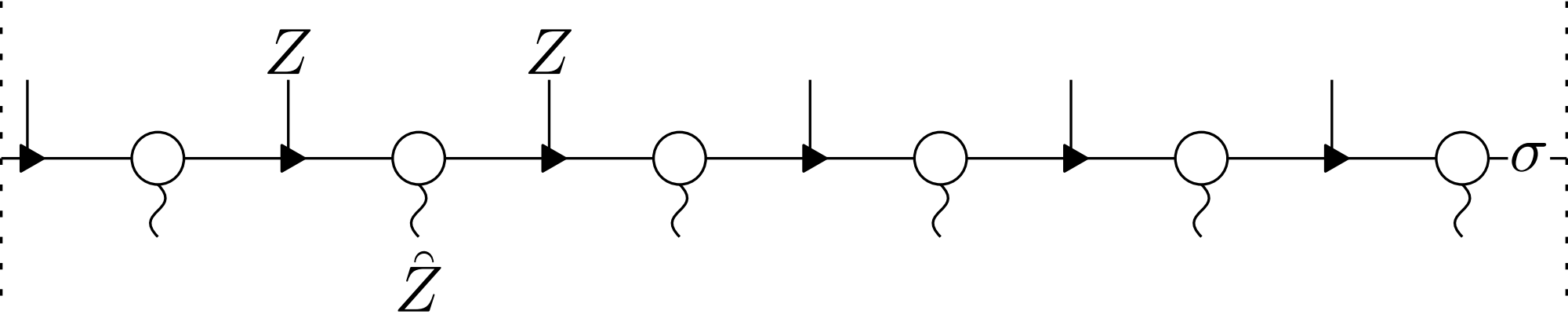

$\hat{Z}$
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$\hat{X}\,\hat{Z}$
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$\hat{X}$

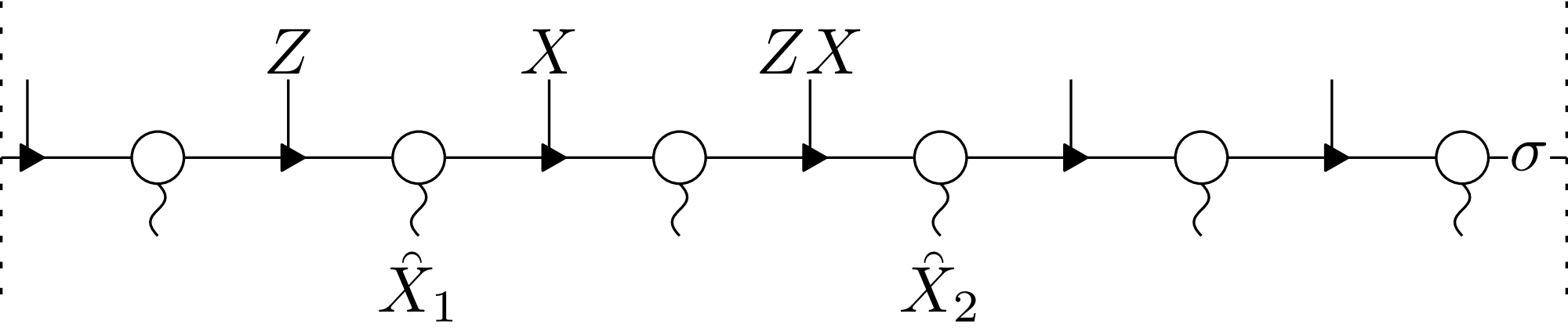

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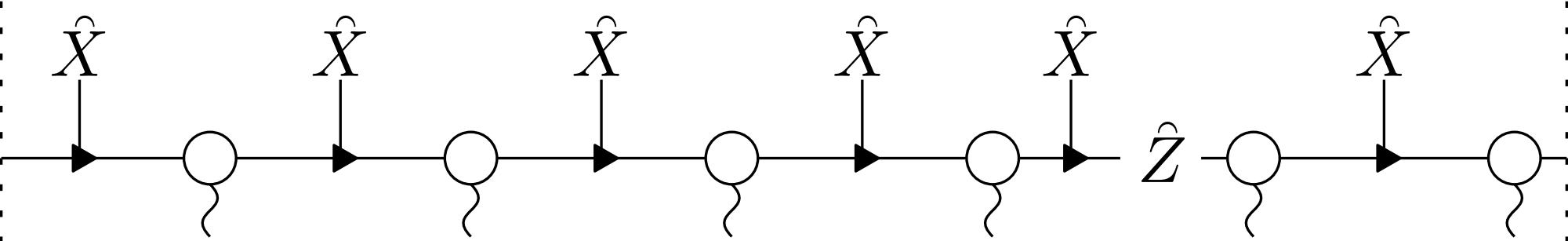

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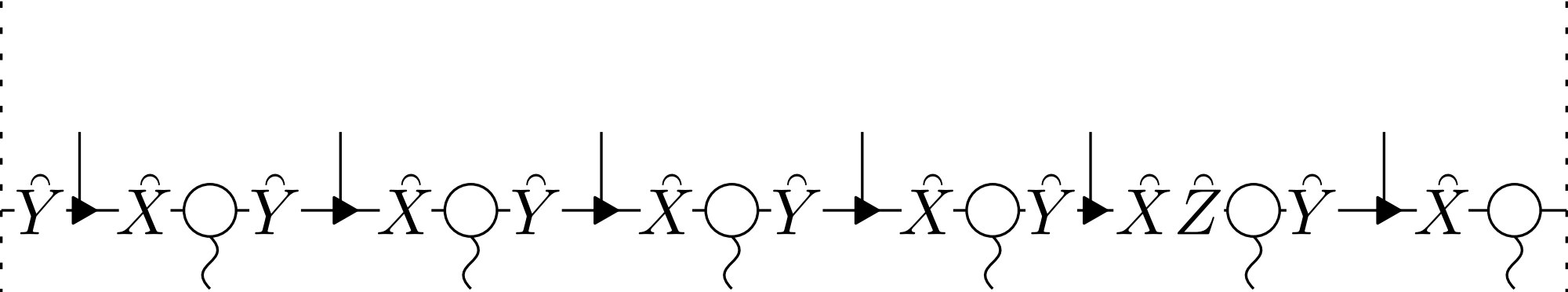

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Z

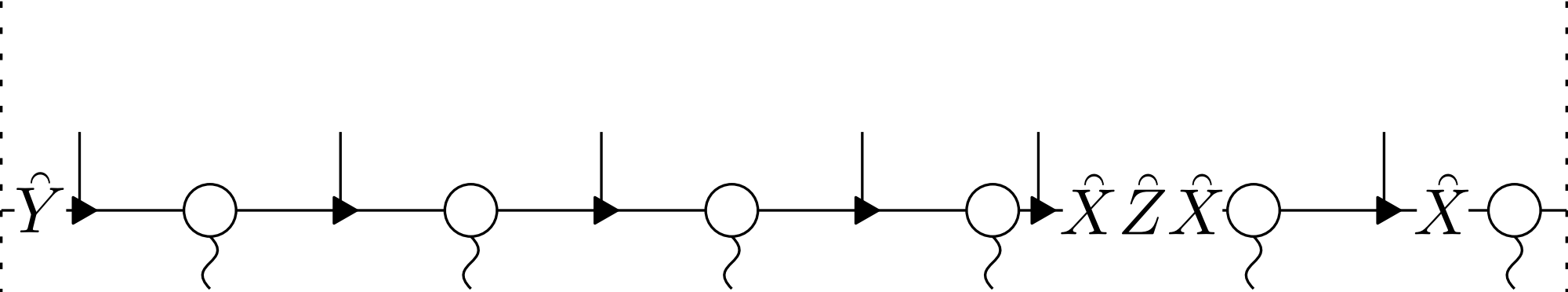

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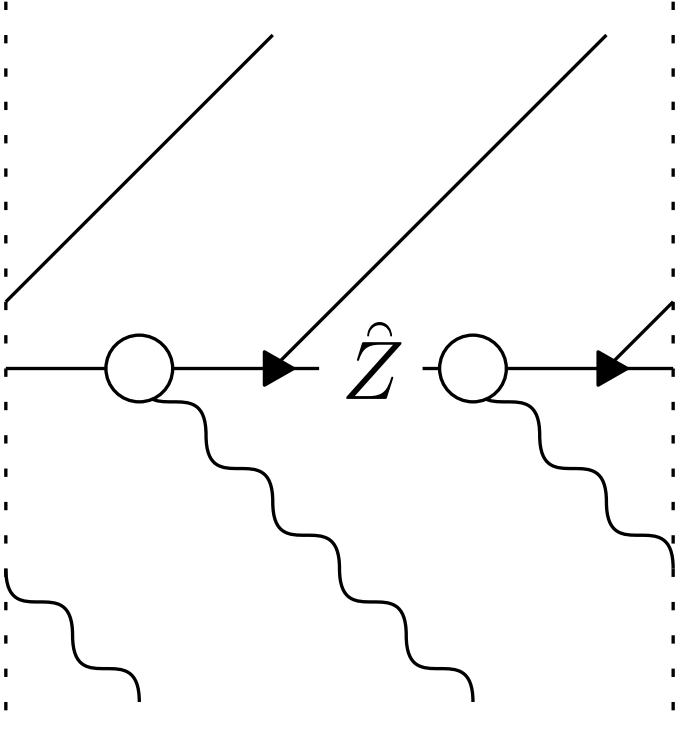

Z
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$\hat{Z}$
σ

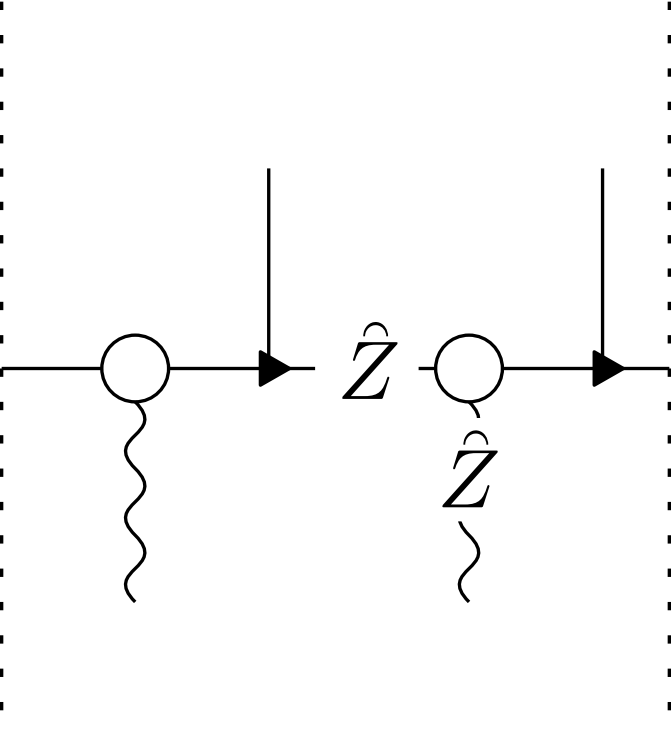

Z
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ZX
σ
X̂1
X̂2

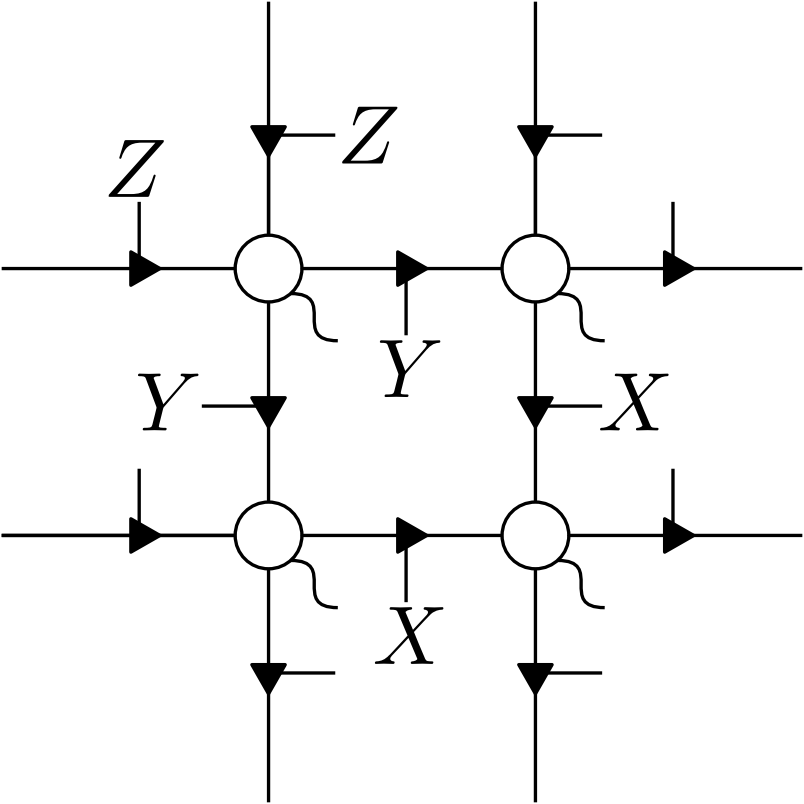

$\hat{X}$
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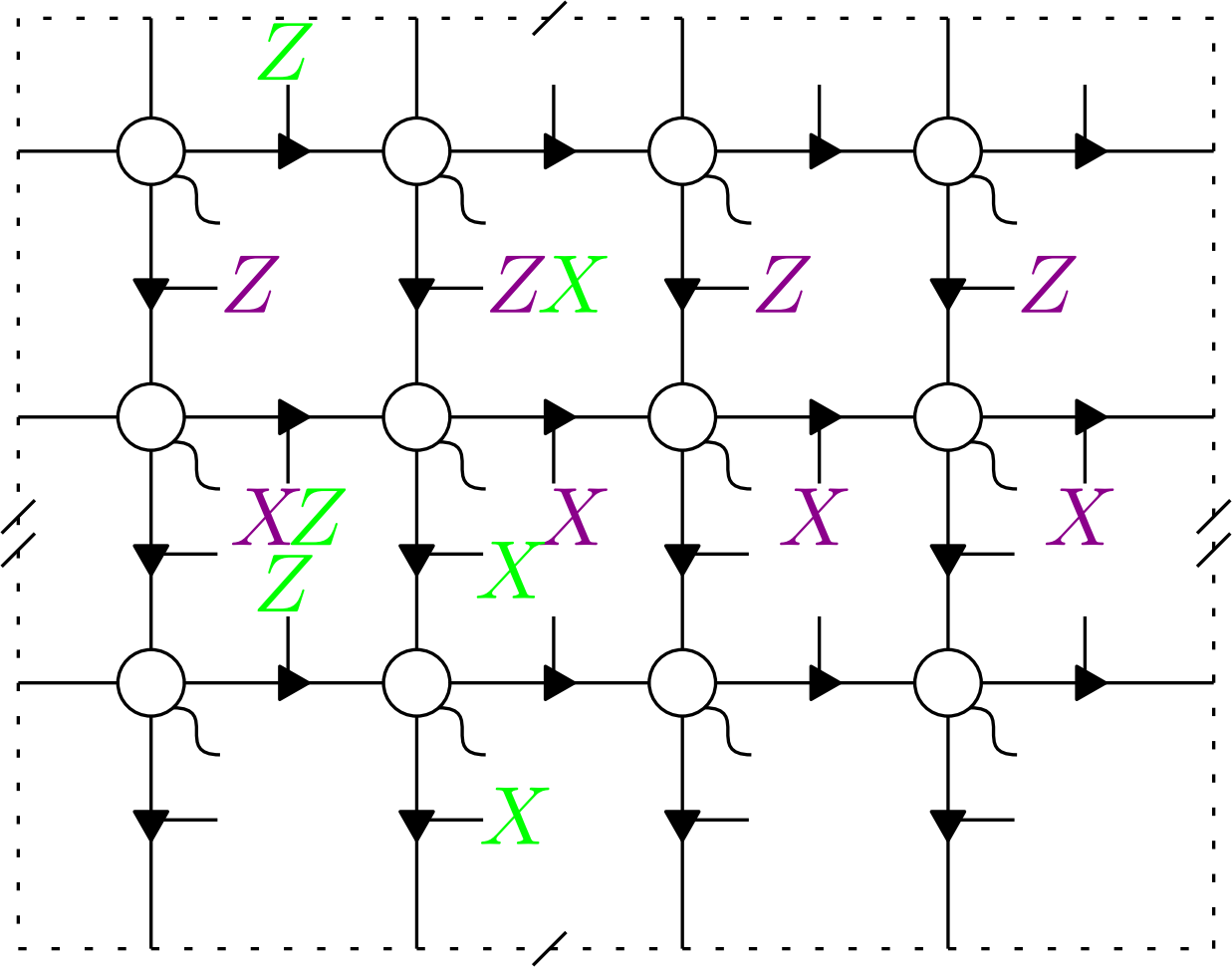

$\hat{Y}$
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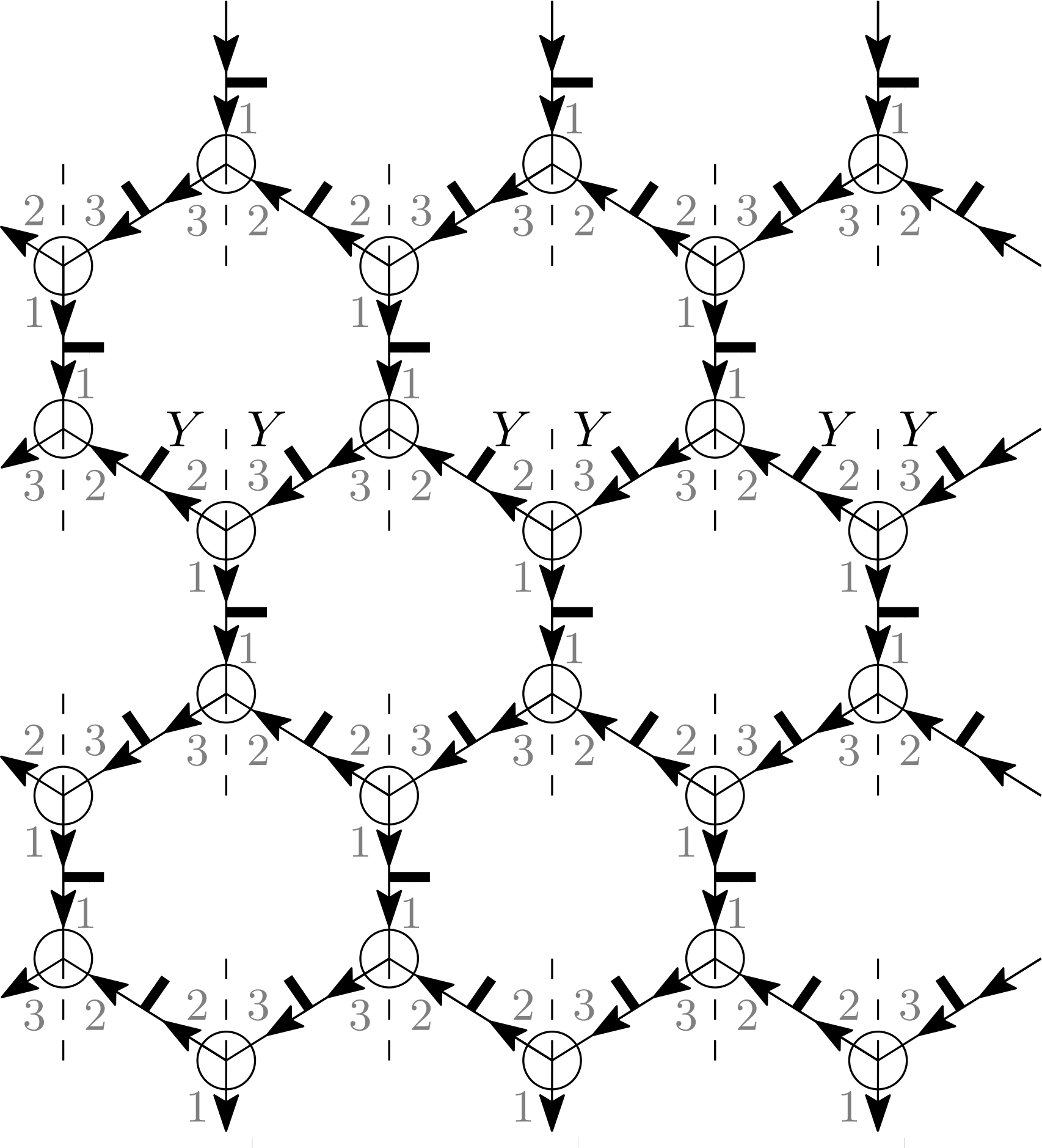

$\hat{Y}$
$\hat{X}\,\hat{Z}\,\hat{X}$
$\hat{X}$

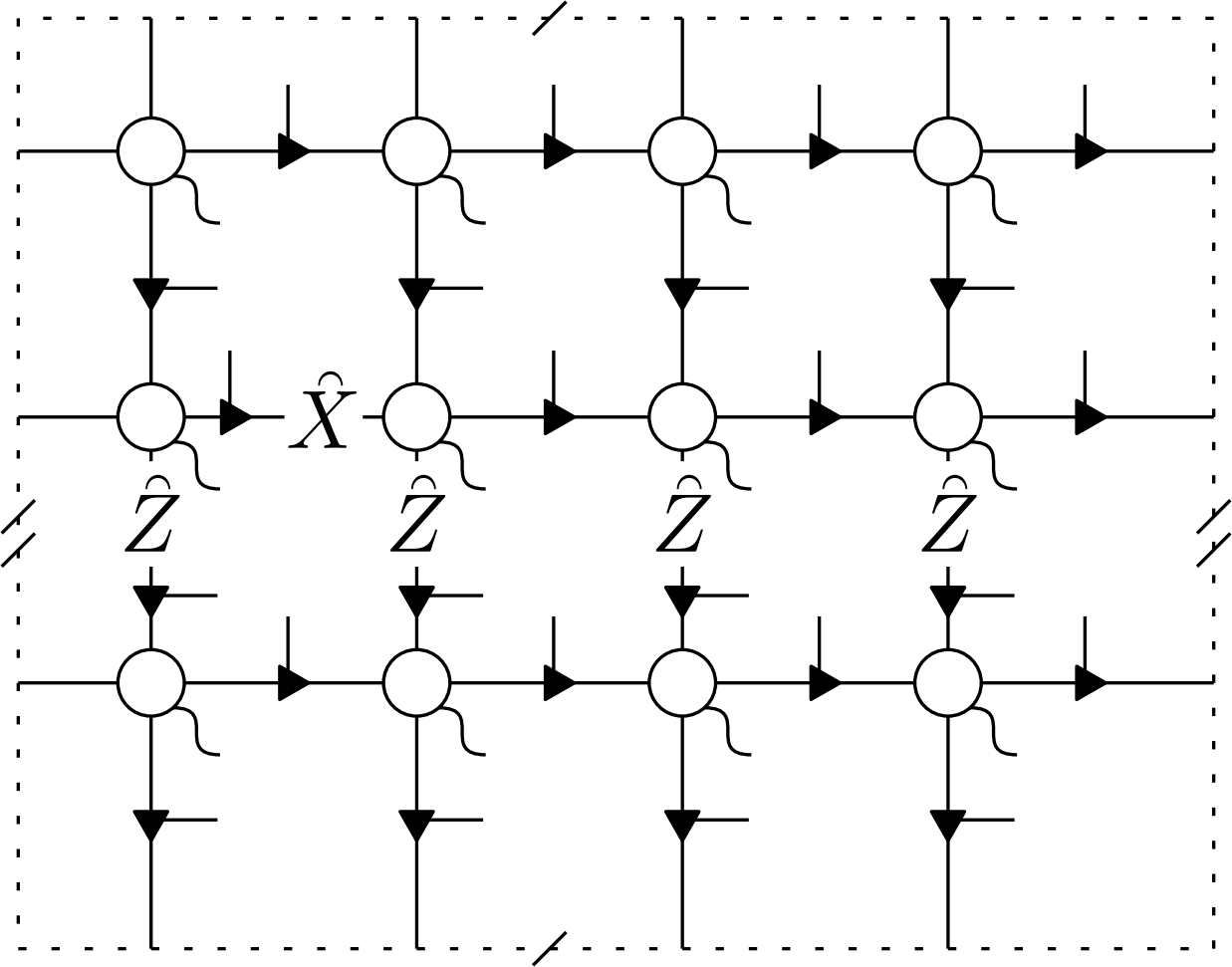

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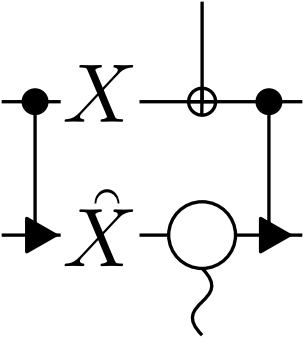

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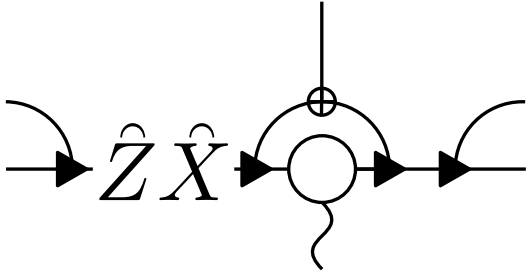

Z
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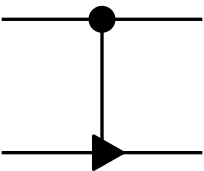

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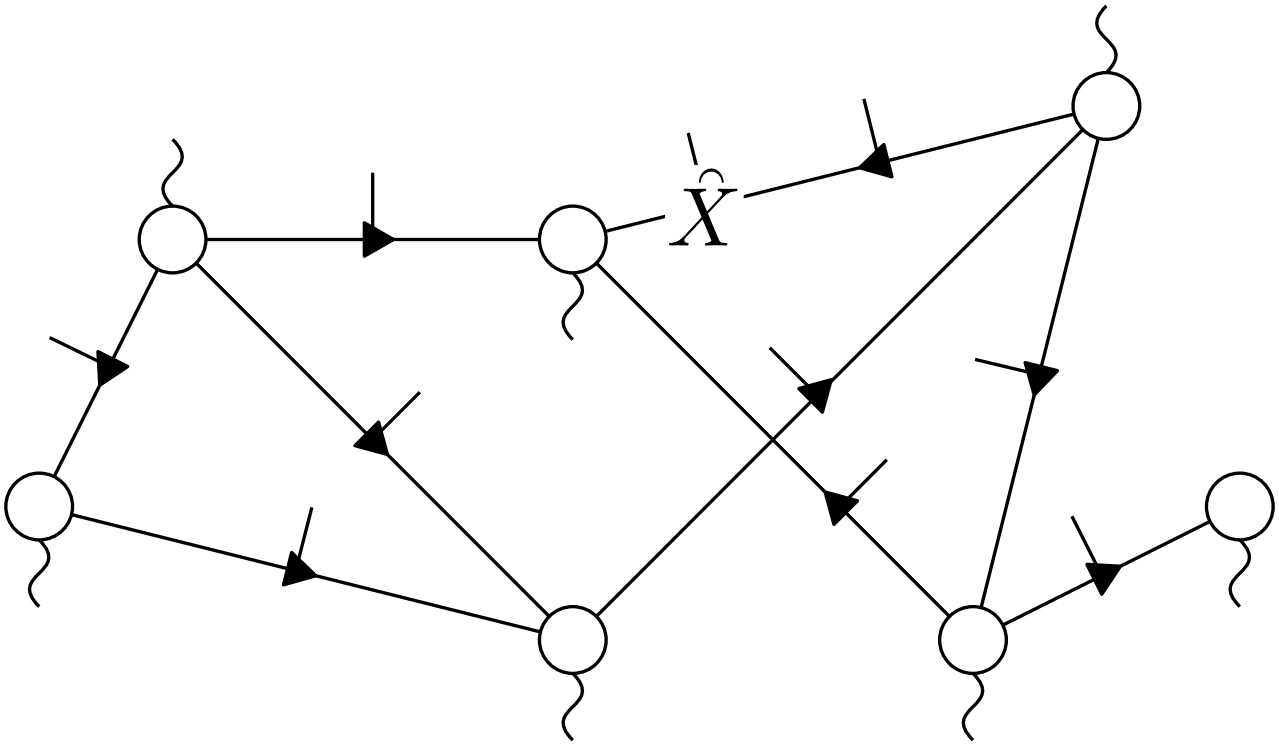

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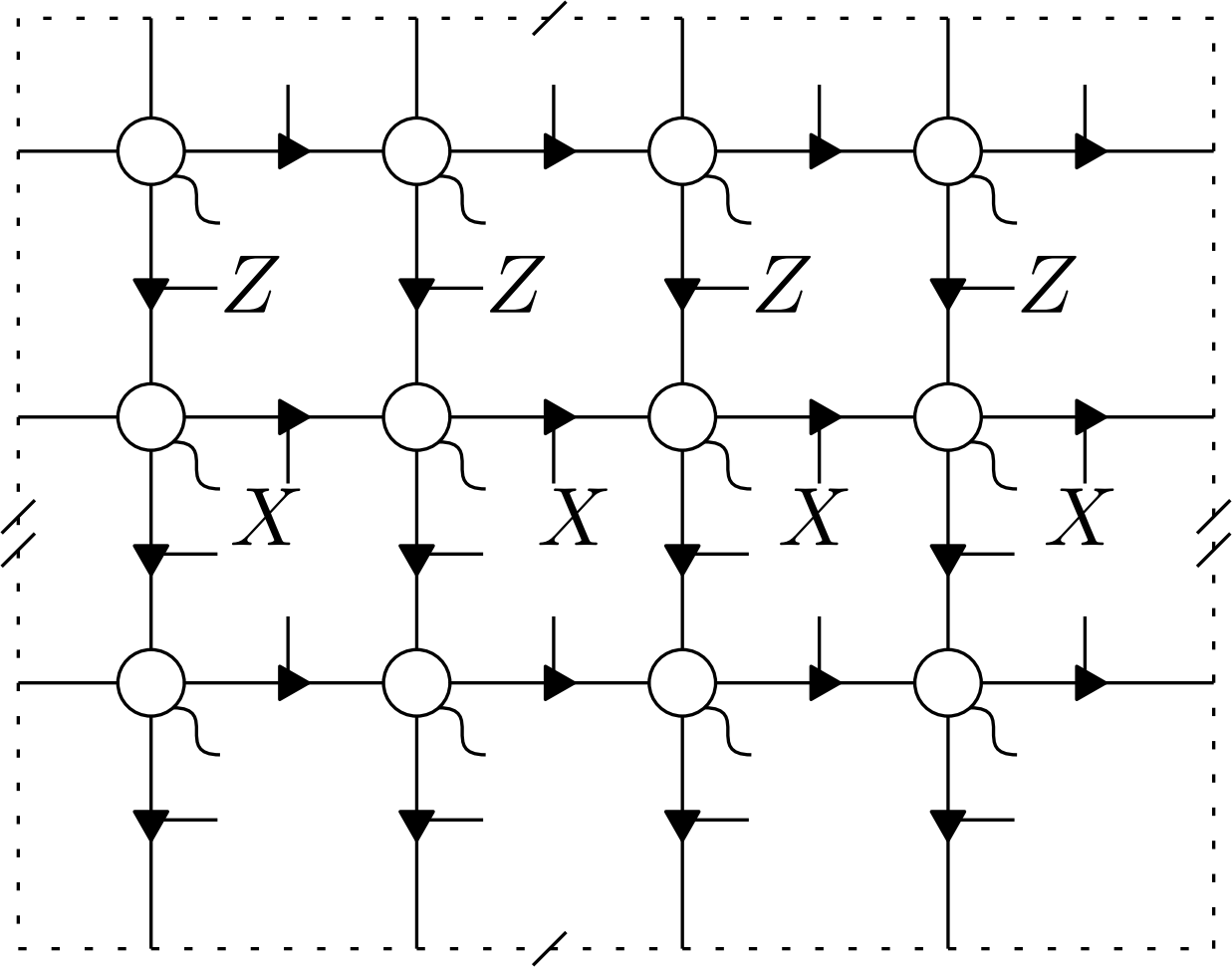

X
X̂

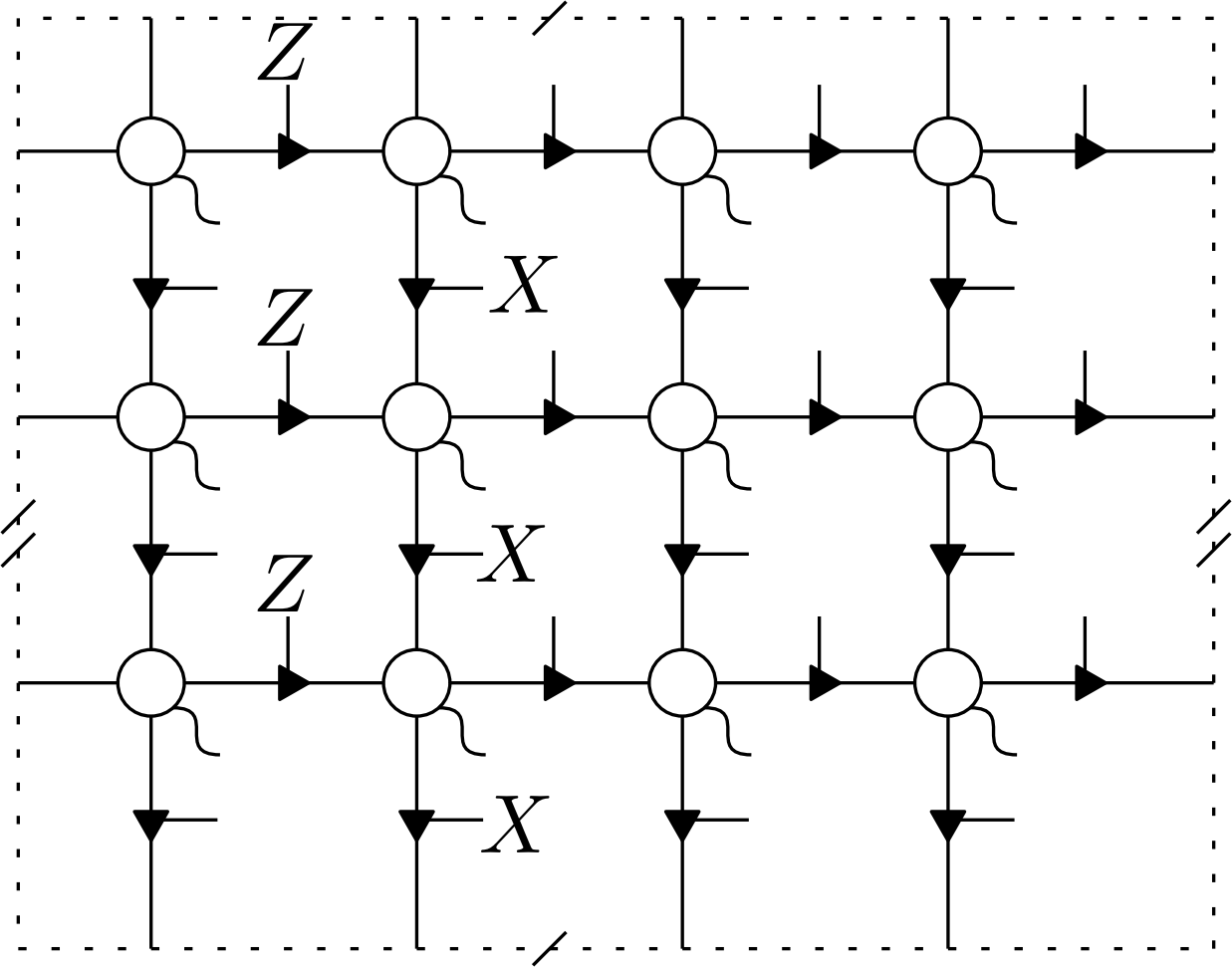

$\hat{Z}\,\hat{X}$

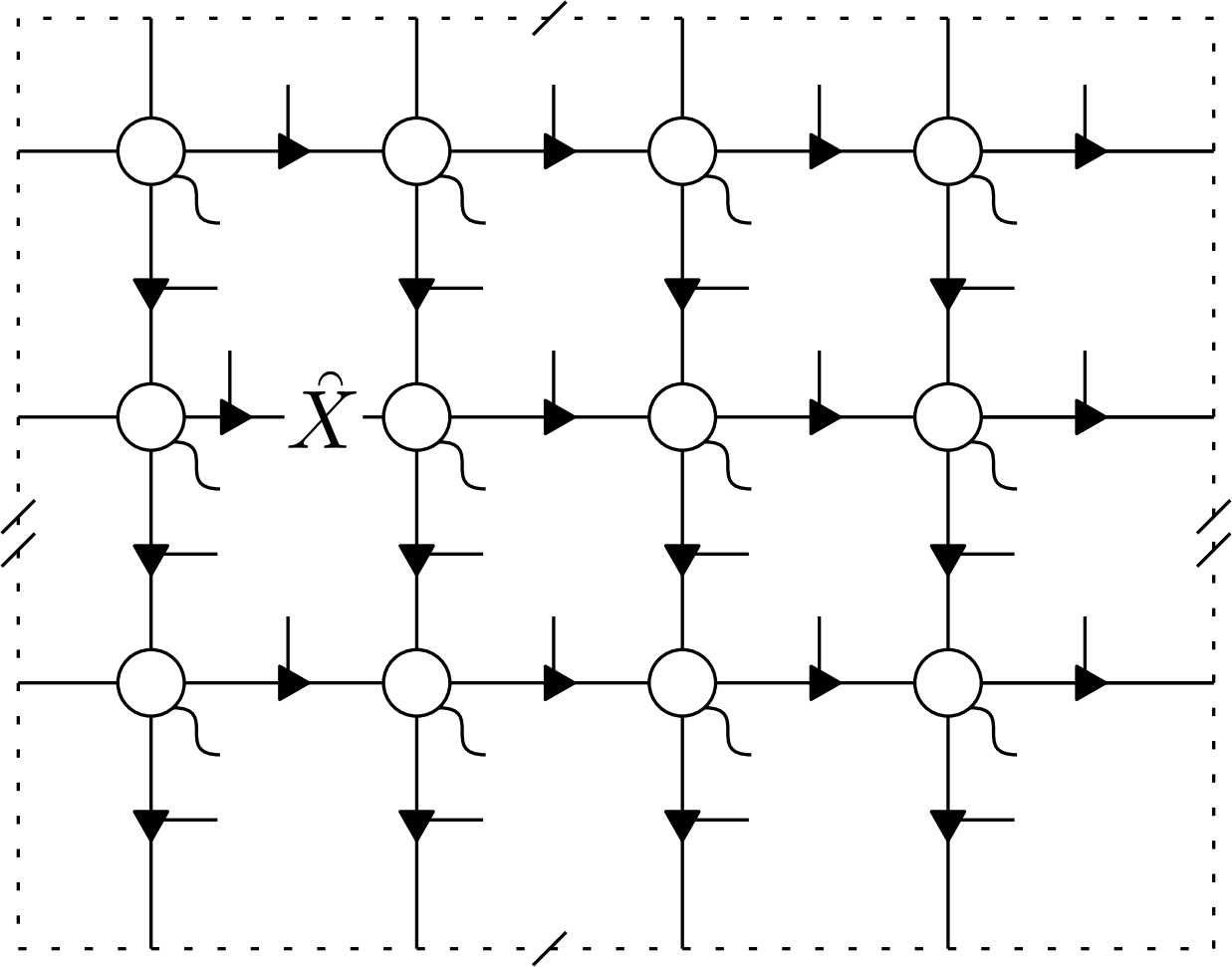


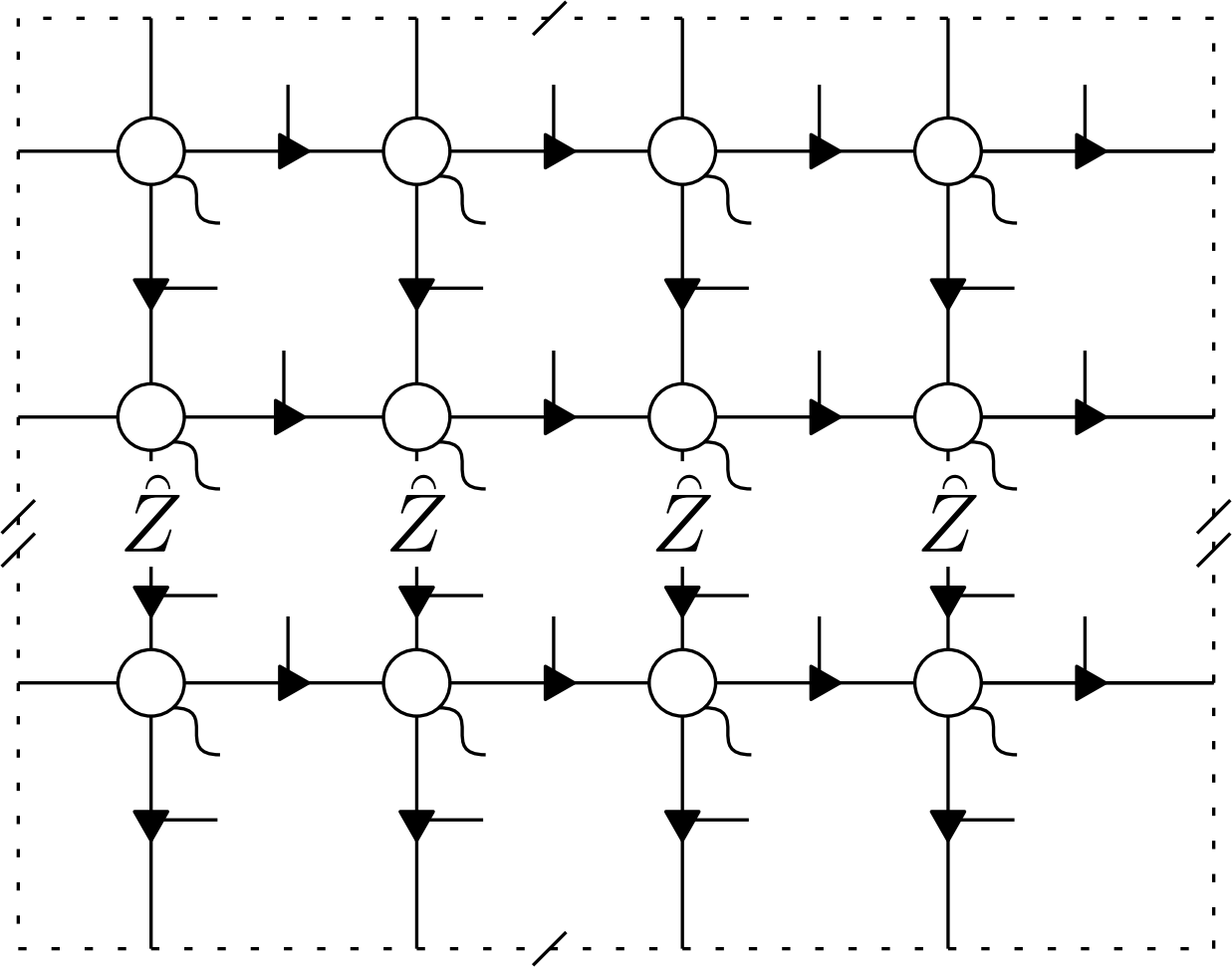


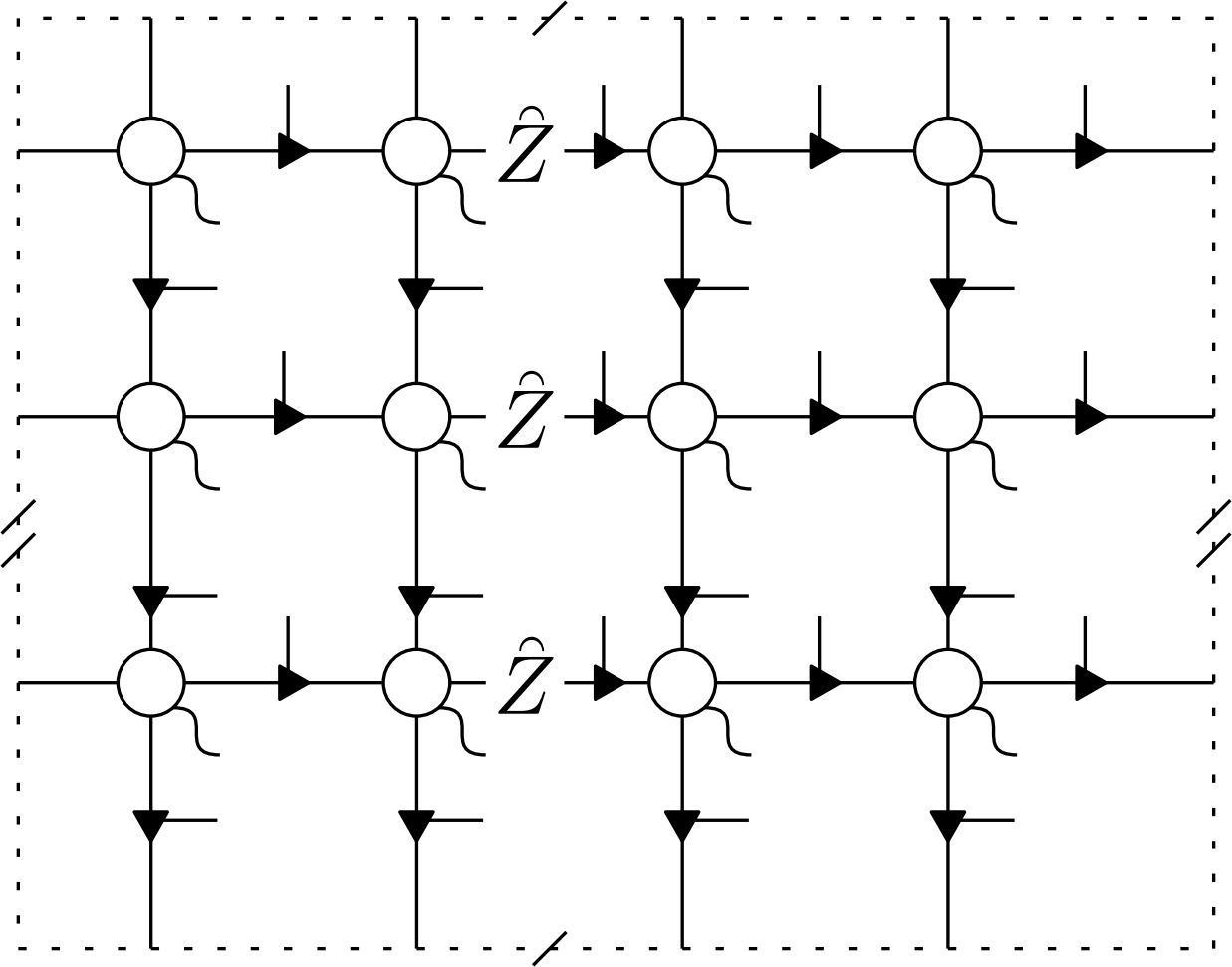

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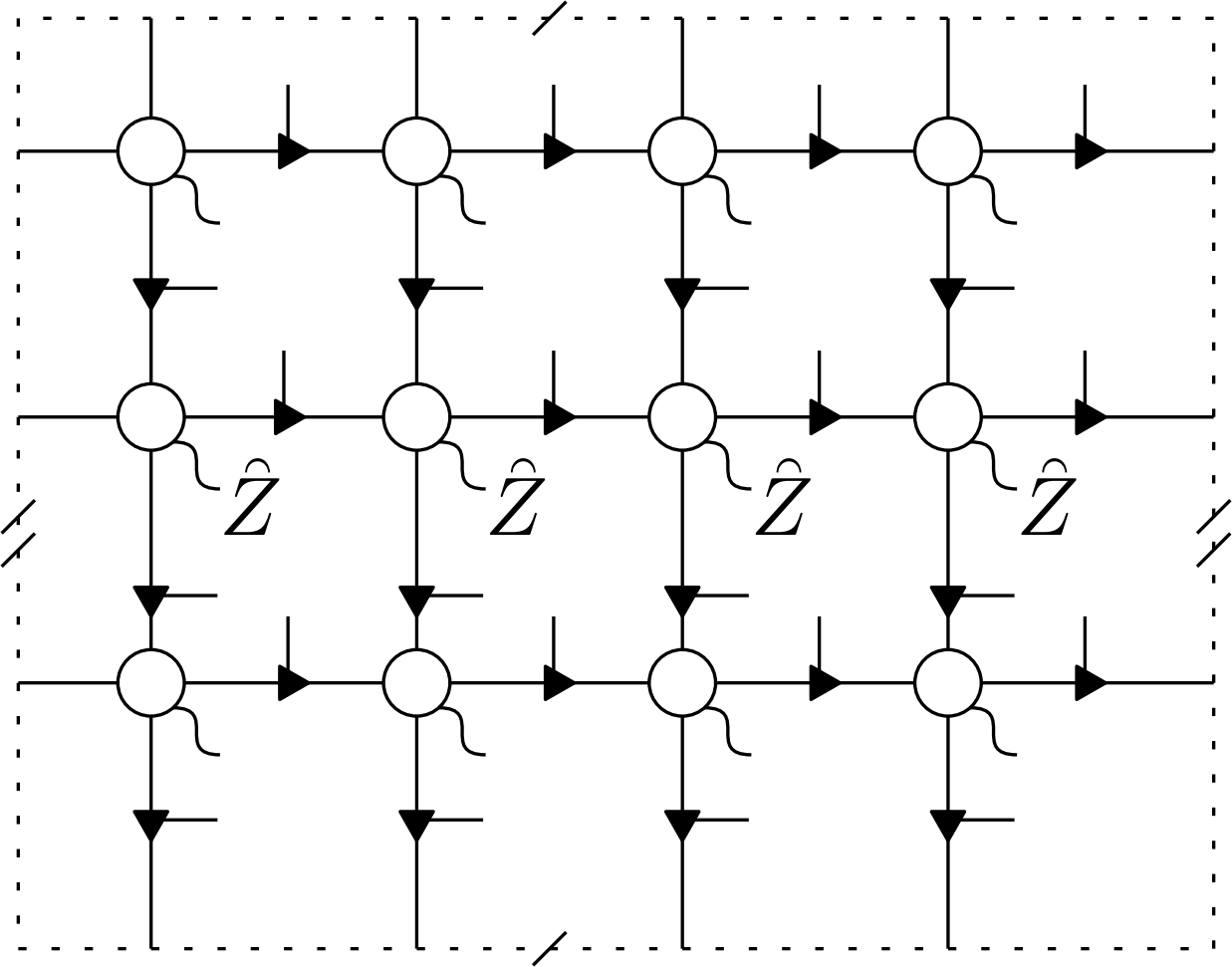

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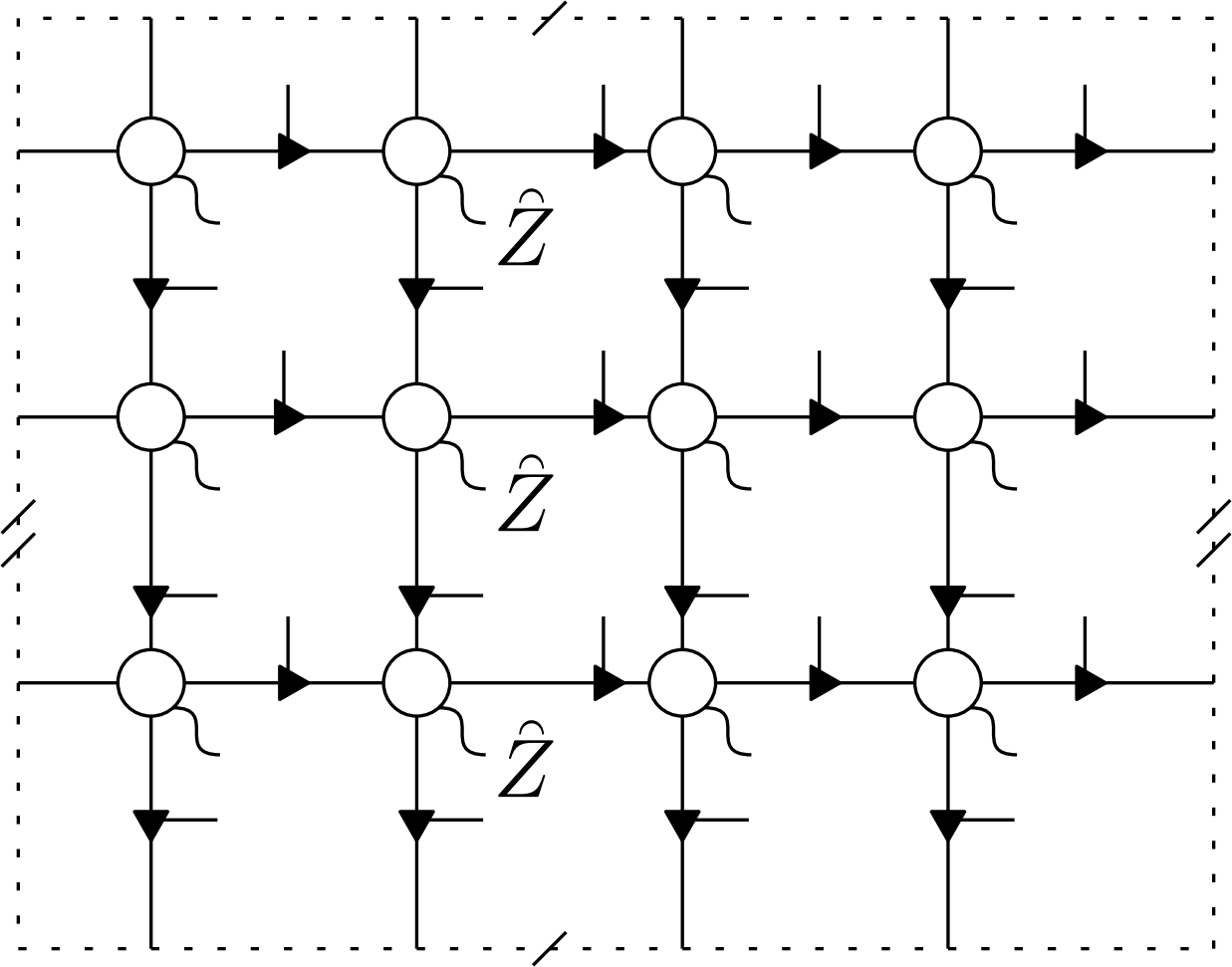

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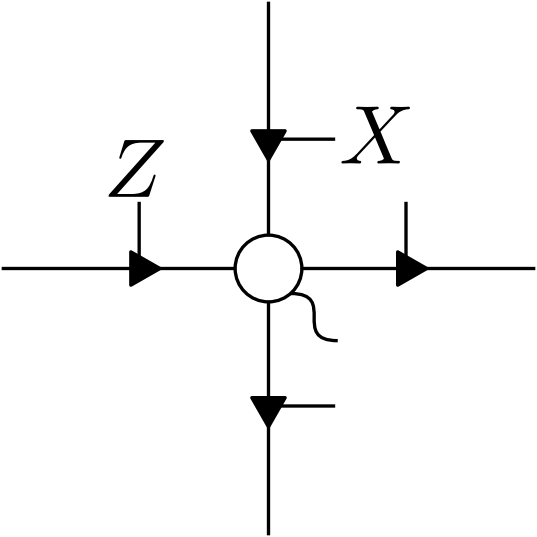

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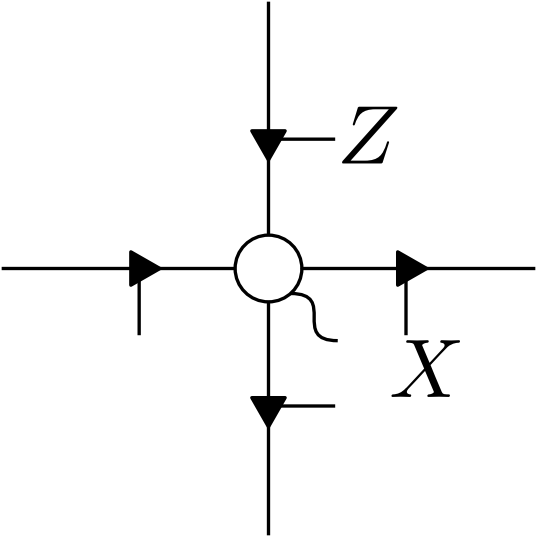

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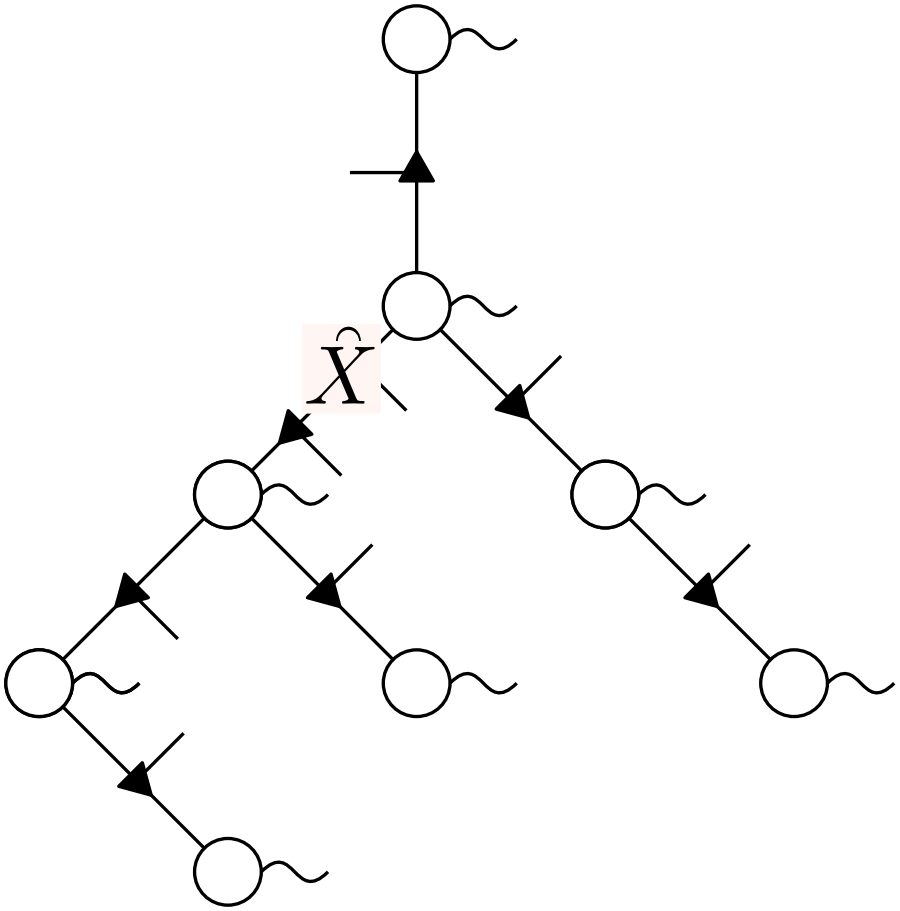

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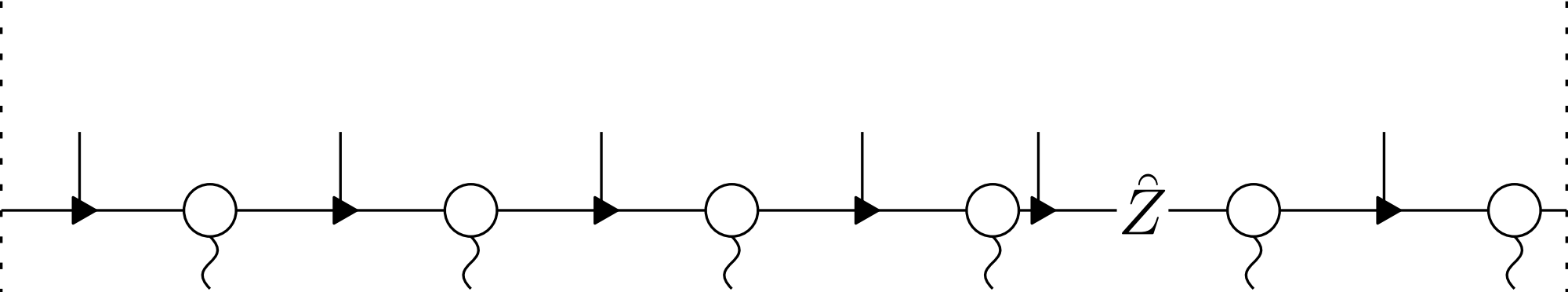

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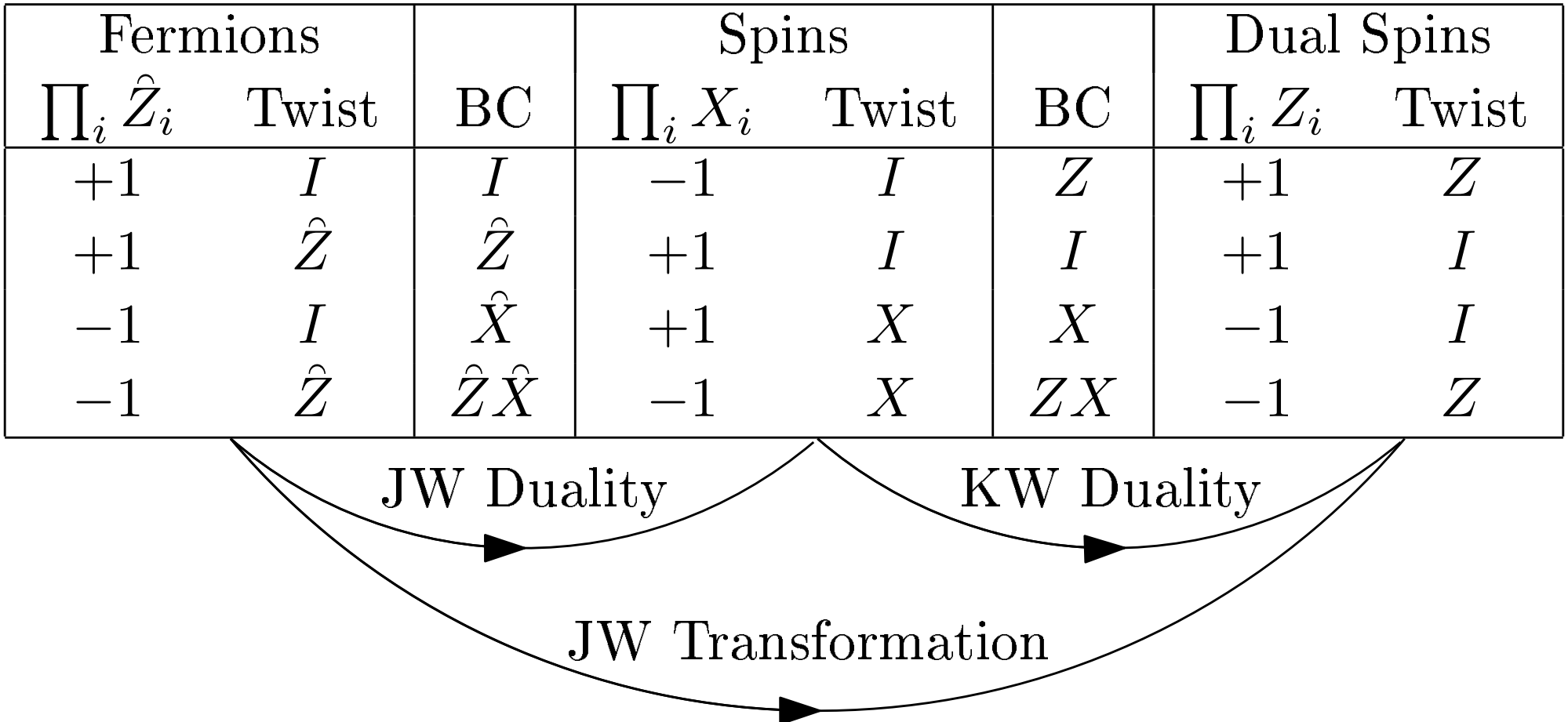

$\hat{Z}$
$\hat{Z}$
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X
Z


Z
X


$\hat{X}$


$\hat{Z}$


Fermions
$\prod_i \hat{Z}_i$
Twist
BC
Spins
$\prod_i X_i$
Twist
BC
Dual Spins
$\prod_i Z_i$
Twist
$+1$ I I -1 I Z $+1$ Z
$+1$ $\hat{Z}$ $\hat{Z}$ $+1$ I I $+1$ I
-1 I $\hat{X}$ $+1$ X X -1 I
-1 $\hat{Z}$ $\hat{Z}\hat{X}$ -1 X ZX -1 Z
JW Duality
KW Duality
JW Transformation

